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International Phonetic Alphabet

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*For the spelling alphabet used in spoken communication over radio, see **NATO phonetic alphabet**.*

*For an introductory guide on IPA symbols with audio, see **Help:IPA**. For the usage of the IPA on Wikipedia, see **Help:IPA/Introduction** and **Help:IPA/English**.*

The **International Phonetic Alphabet** (**IPA**) is an **alphabetic** system of **phonetic notation** based primarily on the **Latin script**. It was devised by the **International Phonetic Association** in the late 19th century as a standard written representation for the sounds of **speech**.^[1] The IPA is used by **linguists**, **lexicographers**, **foreign language** students and teachers, **speech–language pathologists**, singers, actors, **constructed language** creators, and **translators**.^{[2][3]}

The IPA is designed to represent those qualities of **speech** that are part of **lexical** (and, to a limited extent, **prosodic**) sounds in **spoken (oral) language**: **phones**, **intonation** and the separation of **syllables**.^[1] To represent additional qualities of speech – such as tooth **gnashing**, **lisping**, and sounds made with a **cleft palate** – an **extended set of symbols** may be used.^[2]

Segments are **transcribed** by one or more IPA symbols of two basic types: **letters** and **diacritics**. For example, the sound of the English **digraph** (ch) may be transcribed in IPA with a single letter: [ç],^[*not verified in body*] or with multiple letters: [tʃ], or multiple letters plus diacritics: [tʃʰ], depending on how precise one wishes to be. Slashes are used to signal **phonemic transcription**; therefore, /tʃ/ is more abstract than either [tʃʰ] or [ç] and might refer to either, depending on the context and language.^[note 1]

Occasionally, letters or diacritics are added, removed, or modified by the International Phonetic Association. As of the most recent change in 2005,^[4] there are 107 segmental letters, an indefinitely large number of suprasegmental letters, 44 diacritics (not counting composites), and four extra-lexical **prosodic** marks in the IPA. These are illustrated in the current **IPA chart**, posted below in this article and on the International Phonetic Association's website.^[5]

History [edit]

*Main article: **History of the International Phonetic Alphabet***

In 1886 a group of French and English language teachers, led by the French linguist **Paul Passy**, formed what would be known from 1897 onwards as the **International Phonetic Association** (in French, *l'Association phonétique internationale*).^[6] The idea of the alphabet had been suggested to Passy by **Otto Jespersen**. It was developed by Passy along with other members of the association, principally **Daniel Jones**. The original IPA alphabet was based on the **Romic alphabet**, an English **spelling reform** created by **Henry Sweet** that in turn was based on the **Palaeotype alphabet** of **Alexander John Ellis**, itself derived from **Lepsius Standard Alphabet** first used for transcribing Ancient Egyptian into German.

The original intent was to make it usable for other languages the values of the symbols were allowed to vary from language to language.^[note 2] For example, the sound [ʃ] (the *sh* in *shoe*) was originally represented with the letter (c) for English but with (x) for French and German; with German, (c) was used for the [x] sound of *Bach*.^[6] With a growing number of transcribed languages this proved impractical, and in 1888 the values of the letters were made uniform across languages. This would provide the base for all future revisions.^{[6][8]}

Since its creation, the IPA has undergone a number of revisions. After relatively frequent revisions and expansions from the 1890s to the 1940s, the IPA remained nearly static until the **Kiel Convention** in 1989, which substantially revamped the alphabet. A smaller revision took place in 1993 with the resurrection of letters for **mid central vowels**^[2] and the retirement of letters for **voiceless implosives**.^[9] The alphabet was last revised in May 2005 with the addition of a letter for a **labiodental flap**.^[10] Apart from the addition and removal of symbols, changes to the IPA have consisted largely of renaming symbols and categories and in modifying **typefaces**.^[2]

Extensions to the International Phonetic Alphabet for **speech pathology** (extIPA) were created in 1990 and were officially adopted by the **International Clinical Phonetics and Linguistics Association** in 1994.^[11] They were substantially revised in 2015.

Description [edit]

The general principle of the IPA is to provide one letter for each distinctive sound (**speech segment**).^[note 3] This means that:

- It does not normally use **combinations of letters** to represent single sounds, the way English does with ⟨sh⟩, ⟨th⟩ and ⟨ng⟩, nor single letters to represent multiple sounds, the way ⟨x⟩ represents /ks/ or /gz/ in English.
- There are no letters that have context-dependent sound values, the way ⟨c⟩ and ⟨g⟩ in several European languages have a "hard" or "soft" pronunciation.
- The IPA does not usually have separate letters for two sounds if no known language makes a distinction between them, a property known as "selectiveness".^{[2][note 4]} However, if a large number of phonemically distinct letters can be derived with a diacritic, that may be used instead.^[note 5]

The alphabet is designed for transcribing sounds (phones), not **phonemes**, though it is used for phonemic transcription as well. A few letters that did not indicate specific sounds have been retired – (˞), once used for the "compound" tone of Swedish and Norwegian, and ⟨ŋ⟩, once used for the **moraic** nasal of Japanese – though one remains: ⟨ʃ⟩, used for the **sj-sound** of Swedish. When the IPA is used for broad phonetic or for phonemic transcription, the letter–sound correspondence can be rather loose. The IPA has recommended that more 'familiar' letters be used when that would not cause ambiguity.^[13] For example, ⟨e⟩ and ⟨o⟩ for [ɛ] and [ɔ], ⟨t⟩ for [t̪] or [t̚], ⟨f⟩ for [ɸ], etc. Indeed, in the illustration of Hindi in the IPA *Handbook*, the letters ⟨c⟩ and ⟨ɟ⟩ are used for /t͡ʃ/ and /d͡ʒ/.

Among the symbols of the IPA, 107 letters represent **consonants** and **vowels**, 31 **diacritics** are used to modify these, and 17 additional signs indicate **suprasegmental** qualities such as **length**, **tone**, **stress**, and **intonation**.^[note 6] These are organized into a chart; the chart displayed here is the official chart as posted at the website of the IPA.

Letter forms [edit]

The letters chosen for the IPA are meant to harmonize with the **Latin alphabet**.^[note 7] For this reason, most letters are either **Latin** or **Greek**, or modifications thereof. Some letters are neither: for example, the letter denoting the **glottal stop**, ⟨ʔ⟩, originally had the form of a **question mark** with the dot removed. A few letters, such as that of the **voiced pharyngeal fricative**, ⟨ʕ⟩, were inspired by other writing systems (in this case, the **Arabic** letter ⟨ع⟩, ‘ayn, via the reversed apostrophe).^[9]

Some letter forms derive from existing letters:

- The right-swinging tail, as in ⟨t̪ d̪ ɳ ʈ ʂ ʐ ɮ⟩, indicates **retroflex** articulation. It originates from the hook of an *r*.
- The top hook, as in ⟨ɡ ɖ ɓ⟩, indicates **implosion**.
- Several **nasal consonants** are based on the form ⟨n⟩: ⟨ɳ ɲ ɳ ɳ⟩. ⟨ɲ⟩ and ⟨ŋ⟩ derive from **ligatures** of *gn* and *ng*, and ⟨ɱ⟩ is an *ad hoc* imitation of ⟨ŋ⟩.
- Letters turned 180 degrees** for suggestive shapes, such as ⟨e ɔ ə ɓ ɗ ɥ ɹ ɻ ɭ ɮ ɱ ɰ⟩ from ⟨a c e f g h m r v w y⟩.^[note 8] Either the original letter may be reminiscent of the target sound, e.g., ⟨e ə ɹ ɱ⟩ – or the turned one, e.g., ⟨ɔ ɓ ɗ ɥ ɹ ɻ ɭ ɮ ɱ ɰ⟩ recall *o j b y u/w A y/ɹ*. Rotation was popular in the era of **mechanical typesetting**, as it had the advantage of not requiring the casting of special type for IPA symbols, much as the sorts had traditionally often pulled double duty for ⟨b⟩ and ⟨q⟩, ⟨d⟩ and ⟨p⟩, ⟨n⟩ and ⟨u⟩, and ⟨6⟩ and ⟨9⟩ to reduce cost.^[*citation needed*]
- Among consonant letters, the **small capital** letters ⟨G H L N R ʙ⟩, and also ⟨ ⟩ in **extIPA**, indicate more **guttural** sounds than their base letters – ⟨ʙ⟩ is a late exception. Among vowel letters, small capitals indicate **lax vowels**. By 1947, the original small-cap vowel letters ⟨A ʌ ɪ u⟩ had been replaced by ⟨A ʌ ɪ ʊ/ʊ⟩, with only ⟨Y⟩ remaining as a small capital, though later ⟨ɪ⟩ and ⟨æ⟩ would be restored.

Typography and iconicity [edit]

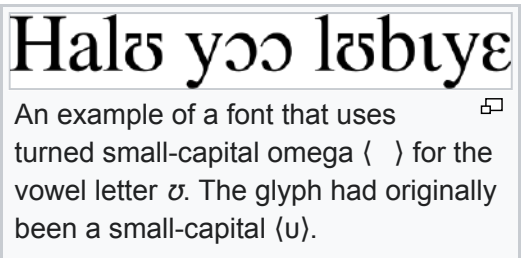
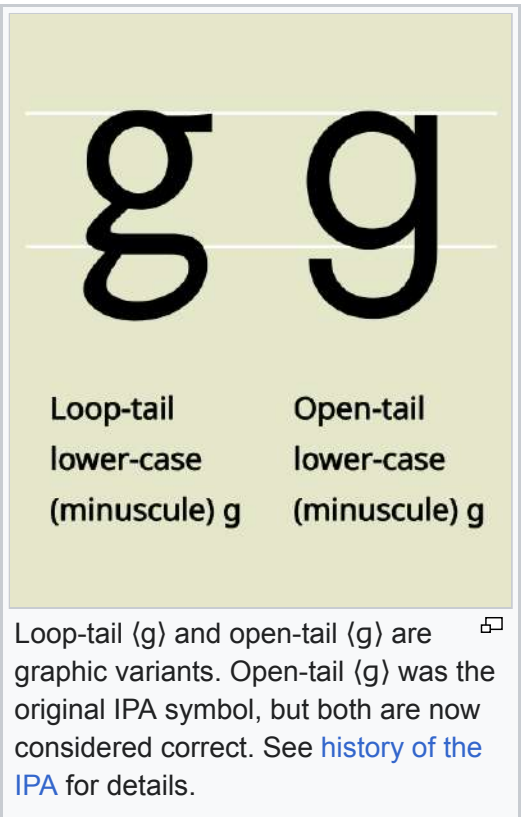
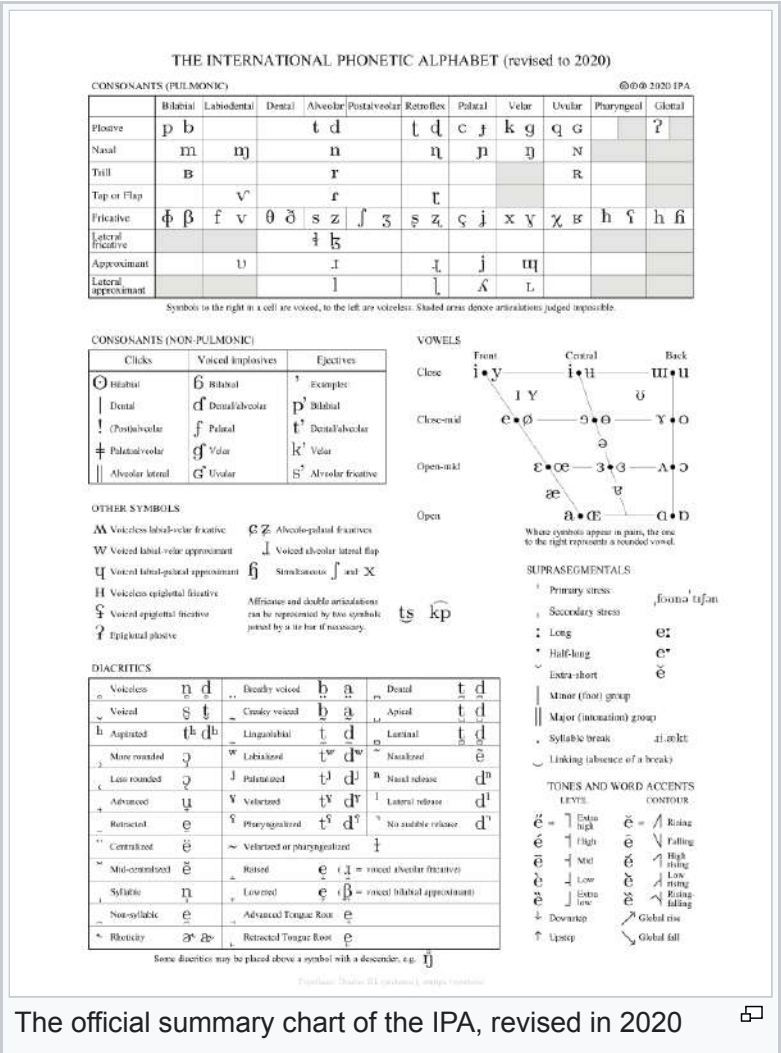
The International Phonetic Alphabet is based on the **Latin script**, and uses as few non-Latin letters as possible.^[6] The Association created the IPA so that the sound values of most letters would correspond to "international usage" (approximately **Classical Latin**).^[6] Hence, the consonant letters ⟨b⟩, ⟨d⟩, ⟨f⟩, ⟨g⟩, ⟨h⟩, ⟨k⟩, ⟨l⟩, ⟨m⟩, ⟨n⟩, ⟨p⟩, ⟨s⟩, ⟨t⟩, ⟨v⟩, ⟨w⟩, and ⟨z⟩ have more or less their word-initial values in English (*g* as in *gill*, *h* as in *hill*, though *p t k* are unaspirated as in *spill*, *still*, *skill*); and the vowel letters ⟨a⟩, ⟨e⟩, ⟨i⟩, ⟨o⟩, ⟨u⟩ correspond to the (long) sound values of Latin: [i] is like the vowel in *machine*, [u] is as in *rule*, etc. Other Latin letters, particularly ⟨j⟩, ⟨r⟩ and ⟨y⟩, differ from English, but have their IPA values in Latin or other European languages.

This basic Latin inventory was extended by adding small-capital and cursive forms, diacritics and rotation. The sound values of these letters are related to those of the original letters, and their derivation may be iconic.^[note 9] For example, letters with a rightward-facing hook at the bottom represent **retroflex** equivalents of the source letters, and small capital letters usually represent **uvular** equivalents of their source letters.

There are also several letters from the Greek alphabet, though their sound values may differ from Greek. For most Greek letters, subtly different **glyph** shapes have been devised for the IPA, specifically ⟨ɑ⟩, ⟨ ⟩, ⟨ɣ⟩, ⟨ɛ⟩, ⟨ɸ⟩, ⟨ ⟩ and ⟨ʊ⟩, which are encoded in **Unicode** separately from their parent Greek letters. One, however – ⟨θ⟩ – has only its Greek form, while for ⟨ ~ β⟩ and ⟨ ~ χ⟩, both Greek and Latin forms are in common use.^[16] The **tone letters** are not derived from an alphabet, but from a **pitch trace** on a **musical scale**.

Beyond the letters themselves, there are a variety of secondary symbols which aid in transcription. **Diacritic** marks can be combined with the letters to add tone and phonetic detail such as **secondary articulation**. There are also special symbols for prosodic features such as stress and intonation.

Brackets and transcription delimiters [edit]



peak may be transcribed as /i/, so that *pick*, *peak* would be transcribed as /ˈpɪk, ˈpiːk/ or as /ˈpɪk, ˈpɪk/; and neither is identical to the vowel of the French *pique*, which would also be transcribed /pik/. By contrast, a narrow phonetic transcription of *pick*, *peak*, *pique* could be: [pʰɪk], [pʰiːk], [pɪkʲ].

Linguists [[edit](#)]

IPA is popular for transcription by linguists. Some American linguists, however, use a mix of IPA with [Americanist phonetic notation](#) or [Sinological phonetic notation](#) or otherwise use [nonstandard symbols](#) for various reasons.^[45] Authors who employ such nonstandard use are encouraged to include a chart or other explanation of their choices, which is good practice in general, as linguists differ in their understanding of the exact meaning of IPA symbols and common conventions change over time.

Dictionaries [[edit](#)]

English [[edit](#)]

Many British dictionaries, including the *Oxford English Dictionary* and some [learner's dictionaries](#) such as the *Oxford Advanced Learner's Dictionary* and the *Cambridge Advanced Learner's Dictionary*, now use [the International Phonetic Alphabet](#) to represent the pronunciation of words.^[46] However, most American (and some British) volumes use one of a variety of [pronunciation respelling](#) systems, intended to be more comfortable for readers of English and to be more acceptable across dialects, without the implication of a preferred pronunciation that the IPA might convey. For example, the respelling systems in many American dictionaries (such as *Merriam-Webster*) use ⟨y⟩ for IPA [j] and ⟨sh⟩ for IPA [ʃ], reflecting the usual spelling of those sounds in English.^{[47][48][note 14]} (In IPA, [y] represents the sound of the French ⟨u⟩, as in *tu*, and [sh] represents the sequence of consonants in *grass**hopper*.)

Other languages [[edit](#)]

The IPA is also not universal among dictionaries in languages other than English. Monolingual dictionaries of languages with [phonemic orthographies](#) generally do not bother with indicating the pronunciation of most words, and tend to use respelling systems for words with unexpected pronunciations. Dictionaries produced in Israel use the IPA rarely and sometimes use the [Hebrew alphabet](#) for transcription of foreign words.^[note 15] Bilingual dictionaries that translate from foreign languages into Russian usually employ the IPA, but monolingual Russian dictionaries occasionally use pronunciation respelling for foreign words.^[note 16] The IPA is more common in bilingual dictionaries, but there are exceptions here too. Mass-market bilingual Czech dictionaries, for instance, tend to use the IPA only for sounds not found in [Czech](#).^[note 17]

Standard orthographies and case variants [[edit](#)]

Main article: [Case variants of IPA letters](#)

IPA letters have been incorporated into the alphabets of various languages, notably via the [Africa Alphabet](#) in many sub-Saharan languages such as [Hausa](#), [Fula](#), [Akan](#), [Gbe languages](#), [Manding languages](#), [Lingala](#), etc. Capital case variants have been created for use in these languages. For example, [Kabiyè](#) of northern [Togo](#) has Ð ɖ, Ɖ ɗ, Ƴ ɣ, Ɔ ɔ, Ɛ ɛ, Ʋ ʋ. These, and others, are supported by [Unicode](#), but appear in Latin ranges other than the [IPA extensions](#).

In the IPA itself, however, only lower-case letters are used. The 1949 edition of the IPA handbook indicated that an asterisk (*) might be prefixed to indicate that a word was a proper name,^[50] and this convention was used by *Le Maître Phonétique*, which was written in IPA rather than in English or French orthography, but it was not included in the 1999 *Handbook*, which notes the contrary use of the asterisk as a placeholder for a sound or feature that does not have a symbol.^[51]

Classical singing [[edit](#)]

The IPA has widespread use among classical singers during preparation as they are frequently required to sing in a variety of foreign languages. They are also taught by vocal coaches to perfect diction and improve tone quality and tuning.^[52] Opera [librettos](#) are authoritatively transcribed in IPA, such as [Nico Castel](#)'s volumes^[53] and Timothy Cheek's book *Singing in Czech*.^[54] Opera singers' ability to read IPA was used by the site *Visual Thesaurus*, which employed several opera singers "to make recordings for the 150,000 words and phrases in VT's lexical database ... for their vocal stamina, attention to the details of enunciation, and most of all, knowledge of IPA".^[55]

Letters [[edit](#)]

See also: [International Phonetic Alphabet chart](#)

The International Phonetic Association organizes the letters of the IPA into three categories: [pulmonic](#) consonants, non-pulmonic consonants, and vowels.^{[note 18][57][58]}

Pulmonic consonant letters are arranged singly or in pairs of voiceless ([tenuis](#)) and voiced sounds, with these then grouped in columns from front (labial) sounds on the left to back (glottal) sounds on the right. In official publications by the IPA, two columns are omitted to save space, with the letters listed among "other symbols" even though theoretically they belong in the main chart.^[note 19] They are arranged in rows from full closure (occlusives: stops and nasals) at top, to brief closure (vibrants: trills and taps), to partial closure (fricatives), and finally minimal closure (approximants) at bottom, again with a row left out to save space. In the table below, a slightly different arrangement is made: All pulmonic consonants are included in the pulmonic-consonant table, and the vibrants and laterals are separated out so that the rows reflect the common [lenition](#) pathway of *stop* → *fricative* → *approximant*, as well as the fact that several letters pull double duty as both fricative and approximant; [affricates](#) may then be created by joining stops and fricatives from adjacent cells. Shaded cells represent articulations that are judged to be impossible or not distinctive.

Vowel letters are also grouped in pairs – of unrounded and rounded vowel sounds – with these pairs also arranged from front on the left to back on the right, and from maximal closure at top to minimal closure at bottom. No vowel letters are omitted from the chart, though in the past some of the mid central vowels were listed among the "other symbols".

Consonants [[edit](#)]

See also: [IPA consonant chart with audio](#)

Pulmonic consonants [[edit](#)]

A [pulmonic](#) consonant is a consonant made by obstructing the [glottis](#) (the space between the vocal folds) or [oral cavity](#) (the mouth) and either simultaneously or subsequently letting out air from the lungs. Pulmonic consonants make up the majority of consonants in the IPA, as well as in human language. All consonants in English fall into this category.^[60]

The pulmonic consonant table, which includes most consonants, is arranged in rows that designate [manner of articulation](#), meaning how the consonant is produced, and columns that designate [place of articulation](#), meaning where in the vocal tract the consonant is produced. The main chart includes only consonants with a single place of articulation.

Place →	Labial			Coronal					Dorsal			Laryngeal	
	Bilabial	Labio-dental	Linguo-labial	Dental	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal/epiglottal	Glottal	
Manner ↓													
Nasal	ᵹ m	ɱ ɱ	ɳ	ɲ n			ɳ̠ ɳ̠	ɲ̟ ɲ̟	ŋ ŋ	ɴ ɴ			
Plosive	p b	p̚ b̚	t̪ d̪	t d			ɖ ɖ	c ɟ	k ɡ	q ɢ	ʕ	ʔ	
Sibilant fricative				s z			ʂ ʐ	ɕ ʑ					
Non-sibilant fricative	ɸ β	f v	θ̪ ð̪	θ ð	ɸ̺ ð̺	ɸ̺̟ ð̺̟	ɸ̺̠ ð̺̠	ç j	x ɣ	χ ʁ	ħ ʕ	h ɦ	
Approximant		ʋ		ɹ			ɻ	j	ɰ			ʎ	
Tap/flap	ɸ̟	ɸ̟̚	ɻ̪	ɾ ɽ			ɽ̠ ɽ̠			ɢ̟	ʕ̟		
Trill	ʙ ɓ			ɽ̺ ɽ̺			ɽ̺̠ ɽ̺̠			ʀ ʁ	ʜ ʕ̼		
Lateral fricative				ɬ ɮ			ɬ̠	ɮ̟	ɮ̠				
Lateral approximant				ɭ			ɭ̠	ʎ̟	ʎ̠	ʎ̟̚			
Lateral tap/flap				ɭ̟ ɭ̟				ʎ̟̚	ʎ̟̚				

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Notes

- In rows where some letters appear in pairs (the *obstruents*), the letter to the right represents a [voiced consonant](#), except [breathy-voiced](#) [ɦ].^[61] In the other rows (the *sonorants*), the single letter represents a voiced consonant.
- While IPA provides a single letter for the coronal places of articulation (for all consonants but fricatives), these do not always have to be used exactly. When dealing with a particular language, the letters may be treated as specifically dental, alveolar, or post-alveolar, as appropriate for that language, without diacritics.
- Shaded areas indicate articulations judged to be impossible.
- The letters [β, ð, ɣ, ʕ, ʕ̼] are canonically voiced fricatives but may be used for approximants.^[note 20]
- In many languages, such as English, [h] and [ɦ] are not actually glottal, fricatives, or approximants. Rather, they are bare [phonation](#).^[63]
- It is primarily the shape of the tongue rather than its position that distinguishes the fricatives [ʐ], [ɕ ʑ], and [ʑ ʐ].
- [ʜ, ʕ̼] are defined as epiglottal fricatives under the "Other symbols" section in the official IPA chart, but they may be treated as trills at the same place of articulation as [ɦ, ʕ̼] because trilling of the [aryepiglottic folds](#) typically co-occurs.^[64]
- Some listed phones are not known to exist as [phonemes](#) in any language.

Non-pulmonic consonants [[edit](#)]

Non-pulmonic consonants are sounds whose airflow is not dependent on the lungs. These include [clicks](#) (found in the [Khoisan languages](#) and some neighboring [Bantu languages](#) of Africa), [implosives](#) (found in languages such as [Sindhi](#), [Hausa](#), [Swahili](#) and [Vietnamese](#)), and [ejectives](#) (found in many [Amerindian](#) and [Caucasian languages](#)).

		ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ	ᄀᄀᄀ
Ejective	Stop	pʼ		tʼ		tʼ	cʼ	kʼ	qʼ	ʔʼ
	Fricative	ɸʼ	fʼ	θʼ	sʼ	ʃʼ	ɕʼ	xʼ	χʼ	

Click (top: velar; bottom: uvular)	Lateral fricative			ɥ					
	Tenuis	k̟ q̟		k q 	k! q!		kɥ qɥ		
	Voiced	ɡ̟ ɠ̟		ɡ ɠ 	ɡ! ɠ!		ɡɥ ɠɥ		
	Nasal	ŋ̟ ɳ̟		ŋ ɳ 	ŋ! ɳ!		ŋɥ ɳɥ		
	Tenuis <i>lateral</i>			k q 					
	Voiced lateral			ɡ ɠ 					
	Nasal lateral			ŋ ɳ 					
Implosive	Voiced	ɓ		ɗ	ɗʼ ɟʼ	ɟ ɠ	ɠ ɡ		
	Voiceless	ɸ		ɗ̥	ɗ̥ʼ ɟ̥ʼ	ɟ̥ ɠ̥	ɠ̥ ɡ̥		

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Notes

- Clicks have traditionally been described as consisting of a forward place of articulation, commonly called the click "type" or historically the "influx", and a rear place of articulation, which when combined with the quality of the click is commonly called the click "accompaniment" or historically the "efflux". The IPA click letters indicate only the click type (forward articulation and release). Therefore, all clicks require two letters for proper notation: ⟨k̠̚, ɡ̠̚, q̠̚⟩, etc., or with the order reversed if both the forward and rear releases are audible. The letter for the rear articulation is frequently omitted, in which case a ⟨k⟩ may usually be assumed. However, some researchers dispute the idea that clicks should be analyzed as doubly articulated, as the traditional transcription implies, and analyze the rear occlusion as solely a part of the airstream mechanism.^[65] In transcriptions of such approaches, the click letter represents both places of articulation, with the different letters representing the different click types, and diacritics are used for the elements of the accompaniment: ⟨l̠, j̠, ʔ̠⟩, etc.
- Letters for the **voiceless** implosives ⟨ɓ, ɸ, ɗ̥, k̥, q̥⟩ are no longer supported by the IPA, though they remain in Unicode. Instead, the IPA typically uses the voiced equivalent with a voiceless diacritic: ⟨ɓ̥, ɟ̥⟩, etc.
- The letter for the **retroflex implosive**, ⟨ɗʼ⟩, is not "explicitly IPA approved",^[66] but the IPA has endorsed the inclusion of ⟨ɗʼ⟩ and voiceless into Unicode.^[*citation needed*]
- The ejective diacritic is placed at the right-hand margin of the consonant, rather than immediately after the letter for the stop: ⟨t͡ʃʼ⟩, ⟨kʷʼ⟩. In imprecise transcription, it often stands in for a superscript glottal stop in **glottalized** but pulmonic **sonorants**, such as [mʔ], [lʔ], [wʔ], [aʔ] – also transcribable as creaky [ɱ̰], [l̰], [ʋ̰], [ə̰].

Affricates [edit]

Affricates and **co-articulated** stops are represented by two letters joined by a tie bar, either above or below the letters with no difference in meaning.^[*note 21*] Affricates are optionally represented by **ligatures** – e.g. ⟨t͡ʃ, d͡ʒ ⟩ – though this is no longer official IPA usage.^[1] Alternatively, a superscript notation for a consonant release is sometimes used to transcribe affricates, for example ⟨t͡ʃʰ⟩ for [t͡ʃs], paralleling [kʰ] ~ [k͡x]. The letters for the palatal plosives ⟨c⟩ and ⟨ɟ⟩ are often used as a convenience for [t͡ʃ] and [d͡ʒ] or similar affricates, even in official IPA publications, so they must be interpreted with care.^[68]

		Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Uvular	Epi-glottal	Glottal
Pulmonic	Sibilant			ts dz		t͡ʃ d͡ʒ	tɕ dʑ	tɕ dʑ				
	Non-sibilant	p͡ɸ b͡ɸ	p͡f b͡ɸ	t̪ d̪	t̺ d̺	t̺ʰ d̺ʰ		c͡ç ɟ͡ʝ	k͡x ɡ͡ɣ	q͡χ ɢ͡ʁ	ʔ͡ɰ ʔ͡ʕ	ʔ͡h
	Lateral			tɬ dɬ			tɭ d	c ɟ͡ʝ	k ɡ͡ɣ			
Ejective	Central			t̪ʰ	tsʰ	t͡ʃʰ	tɕʰ		kxʰ	qχʰ		
	Lateral			tɬʰ				c ʰ	k ʰ			

IPA help • full chart • template

Because in a true affricate the plosive element and the fricative element are homorganic, and the place of articulation of an affricate is most audible in the fricative element, the letter for the former will not always be precisely transcribed where such precision would be redundant. For example, while the English *ch* sound is [t͡ʃ] in close transcription, the diacritic is commonly left off, for [t͡ʃ̥]. Similarly, [t͡ʃ̥s] and [d͡ʒ̥] are more commonly written [t͡ʃ̥s] and [d͡ʒ̥], and in the ligatures there is only a single retroflex hook.

Co-articulated consonants [edit]

Co-articulated consonants are sounds that involve two simultaneous **places of articulation** (are pronounced using two parts of the **vocal tract**). In English, the [w] in "went" is a coarticulated consonant, being pronounced by rounding the lips and raising the back of the tongue. Similar sounds are [ɱ] and [ɥ]. In some languages, plosives can be double-articulated, for example in the name of **Laurent Gbagbo**.

Nasal		Fricative/approximant	
ṁ ṃ	Labial–alveolar	ɹ̠ ɻ̠	Labialized palatal
Ṇ ṇ	Labial–retroflex	ɹ̠ ɻ̠	Labialized velar
ṅ ṅ̃	Labial–velar	ɹ̠ ɻ̠	<i>Sj</i> -sound (variable)
Plosive		Lateral approximant	
ṭ ṭ̃ ḍ ḍ̃	Labial–alveolar	ɹ̠ ɻ̠	Velarized alveolar
ṭ̣ ṭ̣̃ ḍ̣ ḍ̣̃	Labial–retroflex	Implosive	
ḱ ḱ̃ ḡ ḡ̃	Labial–velar	ɹ̠ ɻ̠	Labial–velar
		Ejective	
q̟ʔ	Uvular–epiglottal	ṭ̣ʰ	Labial–alveolar
q̣̰	Labial–uvular		

IPA help • full chart • template

Notes

- [ɹ̠], the **Swedish sj-sound**, is described by the IPA as a "simultaneous [ʃ] and [x]", but it is unlikely such a simultaneous fricative actually exists in any language.^[69]
- Multiple tie bars can be used: ⟨ā̯bc̯⟩ or ⟨ā̯b̯c̯⟩. For instance, a pre-voiced velar affricate may be transcribed as ⟨g̊k̊x̊⟩
- If a diacritic needs to be placed on or under a tie bar, the **combining grapheme joiner** (U+034F) needs to be used, as in [ɸ̊g̊àbd̊ò] 'chewed' (**Margi**). Font support is spotty, however.

With the implosives, authors may not bother to redundantly mark both letters as implosive, but instead write them as less-cluttered ⟨ḡḃ⟩ and even ⟨kḃ⟩.

Vowels [edit]

Main article: *Vowel*

See also: *IPA vowel chart with audio*

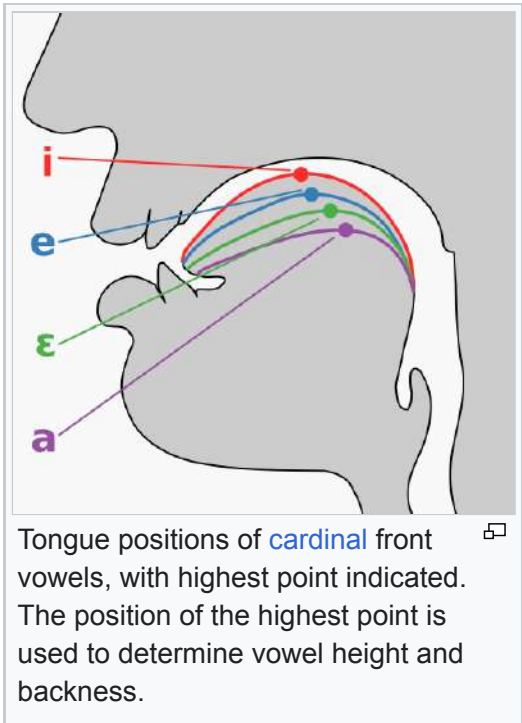
The IPA defines a vowel as a sound which occurs at a syllable center.^[70] Below is a chart depicting the vowels of the IPA. The IPA maps the vowels according to the position of the tongue.

	Front	Central	Back
Close	i • y	ɨ • ʉ	ɯ • u
Near-close	ɪ • ʏ		ʊ
Close-mid	e • ø	ə • ɵ	ɤ • o
Mid	ɛ̞ • ø̞	ə̞	ɤ̞ • o̞
Open-mid	ɛ • œ	ɜ • ɞ	ʌ • ɔ
Near-open	æ	ɐ	
Open	a • ɶ	ä • ɶ̯	ɑ • ɒ

IPA help • audio • full chart • template

The vertical axis of the chart is mapped by **vowel height**. Vowels pronounced with the tongue lowered are at the bottom, and vowels pronounced with the tongue raised are at the top. For example, [ɑ] (the first vowel in *father*) is at the bottom because the tongue is lowered in this position. [i] (the vowel in "meet") is at the top because the sound is said with the tongue raised to the roof of the mouth.

In a similar fashion, the horizontal axis of the chart is determined by **vowel backness**. Vowels with the tongue moved towards the front of the mouth (such as [ɛ], the vowel in "met") are to the left in the chart, while those in which it is moved to the back (such as [ʌ], the vowel in "but") are placed to the right in the chart.



Extensions [\[edit \]](#)

Main article: [Extensions to the International Phonetic Alphabet](#)

The [Extensions to the International Phonetic Alphabet for Disordered Speech](#), commonly abbreviated "extIPA" and sometimes called "Extended IPA", are symbols whose original purpose was to accurately transcribe [disordered speech](#). At the [Kiel Convention](#) in 1989, a group of linguists drew up the initial extensions,^[note 40] which were based on the previous work of the PRDS (Phonetic Representation of Disordered Speech) Group in the early 1980s.^[106] The extensions were first published in 1990, then modified, and published again in 1994 in the *Journal of the International Phonetic Association*, when they were officially adopted by the ICPLA.^[107] While the original purpose was to transcribe disordered speech, linguists have used the extensions to designate a number of sounds within standard communication, such as hushing, gnashing teeth, and smacking lips,^[2] as well as regular lexical sounds such as [lateral fricatives](#) that do not have standard IPA symbols.

In addition to the Extensions to the IPA for disordered speech, there are the conventions of the [Voice Quality Symbols](#), which include a number of symbols for additional airstream mechanisms and secondary articulations in what they call "voice quality".

Associated notation [\[edit \]](#)

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Capital letters and various characters on the number row of the keyboard are commonly used to extend the alphabet in various ways.

Associated symbols [[edit](#)]

There are various punctuation-like conventions for linguistic transcription that are commonly used together with IPA. Some of the more common are:

- (*)
 - (a) A **reconstructed form**.
 - (b) An **ungrammatical form** (including an unphonemic form).

{}**

- (a) A reconstructed form, deeper (more ancient) than a single (*), used when reconstructing even further back from already-starred forms.
- (b) An ungrammatical form. A less common convention than (*) (b), this is sometimes used when reconstructed and ungrammatical forms occur in the same text.^[108]

{x}, {X}

An ungrammatical form. A less common convention than (*) (b), this is sometimes used when reconstructed and ungrammatical forms occur in the same text. [109]

(?)
A doubtfully grammatical form.

 $\langle \% \rangle$

A generalized form, such as a typical shape of a **war**

(#)

A word boundary – e.g. (#V) for a word-initial vowel.

(**\$**) A **phonological word** boundary; e.g. ⟨H\$⟩ for a high tone that occurs in such a position.

(+)
A **morpheme** boundary; e.g. //nɛl+t// for English *kneelt*.

()

The location of a segment – e.g. ⟨V_V⟩ for an intervocalic position, or ⟨ #⟩ for word-final position.

Alternation or contrast^{[citation needed](#)} – e.g. [f] ~ [v] or [f ~ v] for variation between [f] and [v], noting that a /u:/ ~ /ʊ/ contrast is maintained or lost, or indicating the change of a root in e.g. *ni:l* ~ *neɪt* for English *kneel* ~ *knelt*.

(∅)

A null segment or morpheme. This may indicate the absence of an affix, e.g. ⟨kæt-∅⟩ for where an affix might appear but does not (*cat* instead of *cats*), or a deleted segment that leaves a feature behind, such as ⟨∅^w⟩ for an theoretical labialized segment that is only realized as labialization on adjacent segments.^[100]

Capital letters [\[edit \]](#)

Full capital letters are not used as IPA symbols, except as typewriter substitutes (e.g. N for ⟨ŋ⟩, S for ⟨ʃ⟩, O for ⟨ɔ⟩ – see [SAMPa](#)). They are, however, often used in conjunction with the IPA in two cases:^{[*[citation needed](#)*]}

1. for [\(archi\)phonemes](#) and for [natural classes](#) of sounds (that is, as wildcards). The [extIPA](#) chart, for example, uses capital letters as wildcards in its illustrations.
2. as carrying letters for the [Voice Quality Symbols](#).

Wildcards are commonly used in phonology to summarize syllable or word shapes, or to show the evolution of classes of sounds. For example, the possible syllable shapes of Mandarin can be abstracted as ranging from /V/ (an atonic vowel) to /CGVNT/ (a consonant-glide-vowel-nasal syllable with tone), and [word-final devoicing](#) may be schematized as $C \rightarrow \text{C}/_ \#$. They are also used in [historical linguistics](#) for a sound that is posited but whose nature has not been determined beyond some generic category such as {nasal} or {uvular}. In speech pathology, capital letters represent indeterminate sounds, and may be superscripted to indicate they are weakly articulated: e.g. [D^h] is a weak indeterminate alveolar, [K^h] a weak indeterminate velar.^[111]

There is a degree of variation between authors as to the capital letters used, but these are ubiquitous in English-language material:^[112]

- ⟨C⟩ for {consonant}
- ⟨V⟩ for {vowel}
- ⟨N⟩ for {nasal}

Other common conventions are:^[113]

- ⟨T⟩ for {tone/accnt} (tonicity)
- ⟨P⟩ for {plosive}
- ⟨F⟩ for {fricative}
- ⟨S⟩ for {sibilant}^[note 41]
- ⟨G⟩ for {glide/semivowel}
- ⟨L⟩ for {lateral} or {liquid}
- ⟨R⟩ for {rhotic} or {resonant/sonorant}^[note 42]
- ⟨ℳ⟩ for {obstruent}
- ⟨ ˀ ⟩ for {click}
- ⟨A, E, ɪ, O, U⟩ for {open, front, close, back, rounded vowel}^[note 43] and ⟨B, D, ʝ, K, Q, Φ, H⟩ for {labial, alveolar, post-alveolar/palatal, velar, uvular, pharyngeal, glottal^[note 44] consonant}, respectively
- ⟨X⟩ for {any sound}, as in ⟨CVX⟩ for a heavy syllable {CVC, CVV, CV:}

The letters can be modified with IPA diacritics, for example:^[113]

- ⟨C'⟩ for {ejective}
- ⟨C̣⟩ for {implosive}
- ⟨N̥C⟩ or ⟨^NC⟩ for {prenasalized consonant}
- ⟨Ṽ⟩ for {nasal vowel}
- ⟨C^hV̌⟩ for {aspirated CV syllable with high tone}
- ⟨ʒ⟩ for {voiced sibilant}
- ⟨N̥⟩ for {voiceless nasal}
- ⟨P̥F̥⟩ or ⟨P̥ ̥⟩ for {affricate}
- ⟨C^G⟩ for a consonant with a glide as secondary articulation (e.g. ⟨Cᵢ⟩ for {palatalized consonant} or ⟨C^w⟩ for {labialized consonant})
- ⟨D̪⟩ for {dental consonant}

(H), (M), (L) are also commonly used for high, mid and low tone, with (LH) for rising tone and (HL) for falling tone, rather than transcribing them overly precisely with IPA tone letters or with ambiguous digits.^[note 45] When distinguishing five levels of pitch, (xH) and (xL) may be used for 'extra high' and 'extra low'.^[citation needed] Arbitrary sequences of letters such as (A B C D) may be used for tone phonemes, especially when comparing across related languages.

Typical examples of archiphonemic use of capital letters are:^[*citation needed*]

- (I) for the Turkish harmonic vowel set {i y u u} ^[note 46]
- (D) for the conflated flapped middle consonant of American English *writer* and *rider*
- (N) for the [homorganic](#) syllable-coda nasal of languages such as Spanish and Japanese (essentially equivalent to the wild-card usage of the letter)
- (R) in cases where a phonemic distinction between trill /r/ and flap /ɾ/ is conflated, as in Spanish *enrejar* /eNreˈxaR/ (the *n* is homorganic and the first *r* is a trill, but the second *r* is variable).^[14]

Similar usage is found for *phonemic* analysis, where a language does not distinguish sounds that have separate letters in the IPA. For instance, [Castillian Spanish](#) has been analyzed as having phonemes /θ/ and /s/, which surface as [θ] and [s] in voiceless environments and as [ð] and [z] in voiced environments (e.g. *hazte* /'aθe/ → ['aθe], vs *hazme* /'aθme/ → ['aðme], or *las manos* /laS

[illegible]

ASCII and keyboard transliterations [edit]

Further information: *Comparison of ASCII encodings of the International Phonetic Alphabet*

Several systems have been developed that map the IPA symbols to **ASCII** characters. Notable systems include **SAMPA** and **X-SAMPA**. The usage of mapping systems in on-line text has to some extent been adopted in the context input methods, allowing convenient keying of IPA characters that would be otherwise unavailable on standard keyboard layouts.

IETF language tags [edit]

IETF language tags have registered *fonipa* as a variant subtag identifying text as written in IPA.^[129] Thus, an IPA transcription of English could be tagged as *en-fonipa*. For the use of IPA without attribution to a concrete language, *und-fonipa* is available.

Computer input using on-screen keyboard [edit]

Online IPA keyboard utilities are available, though none of them cover the complete range of IPA symbols and diacritics. Examples are the *IPA 2018 i-charts* hosted by the IPA,^[130] *IPA character picker* by Richard Ishida at GitHub,^[131] *Type IPA phonetic symbols* at Typelt.org,^[132] and an *IPA Chart keyboard* by Weston Ruter also at GitHub.^[133] In April 2019, Google's **Gboard** for **Android** added an IPA keyboard to its platform.^{[134][135]} For iOS there are multiple free keyboard layouts available, such as the *IPA Phonetic Keyboard*.^[136]

See also [edit]

- Afroasiatic phonetic notation** – Phonetic notation
- Americanist phonetic notation** – Phonetic alphabet developed in the 1880s
- Articulatory phonetics** – A branch of linguistics studying how humans make sounds
- Case variants of IPA letters** – International Phonetic Alphabet variants
- Cursive forms of the International Phonetic Alphabet** – Deprecated cursive forms of IPA symbols
- Extensions to the International Phonetic Alphabet** – Disordered speech additions to the phonetic alphabet
- Index of phonetics articles**
- International Alphabet of Sanskrit Transliteration** – Transliteration scheme for Indic scripts
- Sound correspondences between English accents**
- List of international common standards**

Notes [edit]

- ↑ The inverted bridge minus under the ⟨ɸ^h⟩ specifies it as **apical** (pronounced with the tip of the tongue), the small minus specifies it as **postalveolar** (pronounced slightly back from the **alveolar ridge**), and the superscript *h* shows that it is **aspirated** (breathy). These qualities cause the English /tʃ/ to sound different from the Italian or Spanish /tʃ/, which is a **laminal** (pronounced with the blade of the tongue)^[*citation needed*] and unaspirated ⟨ɸ^h⟩. ⟨ɸ^h⟩ and ⟨ɸ^h⟩ are thus two different, though similar, sounds.
- ↑ "Originally, the aim was to make available a set of phonetic symbols which would be given *different* articulatory values, if necessary, in different languages."^[7]
- ↑ "From its earliest days [...] the International Phonetic Association has aimed to provide 'a separate sign for each distinctive sound; that is, for each sound which, being used instead of another, in the same language, can change the meaning of a word'."^[12]
- ↑ For instance, **flaps and taps** are two different kinds of **articulation**, but since no language has (yet) been found to make a distinction between, say, an **alveolar flap** and an alveolar tap, the IPA does not provide such sounds with dedicated letters. Instead, it provides a single letter – in this case, [ɾ] – for both. Strictly speaking, this makes the IPA a partially **phonemic** alphabet, not a purely **phonetic** one.
- ↑ This exception to the rules was made primarily to explain why the IPA does not make a dental–alveolar distinction, despite one being phonemic in hundreds of languages, including most of the continent of Australia. **Americanist Phonetic Notation** makes (or at least made) a distinction between apical (ʈ d s z ɳ ɭ) and laminal (ɖ ɟ ʑ ʎ ʟ), which is easily applicable to alveolar vs dental (when a language distinguishes apical alveolar from laminal dental, as in Australia), but despite several proposals to the Council, the IPA never voted to accept such a distinction.
- ↑ There are three basic tone diacritics and five basic tone letters, both sets of which may be compounded.
- ↑ "The non-roman letters of the International Phonetic Alphabet have been designed as far as possible to harmonize well with the roman letters. The Association does not recognize makeshift letters; It recognizes only letters which have been carefully cut so as to be in harmony with the other letters."^[14]
- ↑ Originally, ⟨ʊ⟩ was written as a small capital U. However, this was not easy to read, and so it was replaced with a turned small capital omega. In modern typefaces, it often has its own design, called a "horseshoe".
- ↑ "The new letters should be suggestive of the sounds they represent, by their resemblance to the old ones."^[15]
- ↑ **^{*a*}****^{*b*}** For example, **Merriam-Webster** dictionaries use **backslashes** \\ ... \\ to demarcate their in-house diaphonemic transcription system. This contrasts with the *Oxford English Dictionary*, which transcribes a specific target accent.
- ↑ For example, single and double pipe symbols are used for minor and major prosodic breaks. Although the *Handbook* specifies the prosodic symbols as being "thick" vertical lines, which would in theory be distinct from simple ASCII pipes used as delimiters (and similar to *Dania* transcription), this was an idea to keep them distinct from the otherwise similar pipes used as **click letters**, and is almost never found in practice.^[26] The *Handbook* assigns the prosodic pipe the Unicode encodings U+007C, which is the simple ASCII symbol, and the double pipe U+2016.^[27]
- ↑ The proper angle brackets in Unicode are the mathematical symbols (U+27E8 and U+27E9). Chevrons <...> (U+2039, U+203A) are sometimes substituted, as in Americanist phonetic notation, as are the less-than and greater-than signs <...> (U+003C, U+003E) found on ASCII keyboards.
- ↑ See "Illustrations of the IPA" in the *Handbook* for individual languages which for example may use /c/ as a phonemic symbol for what is phonetically realized as [tʃ], or superscript variants of IPA letters that are not officially defined.
- ↑ ***Pronunciation respelling for English*** contains detailed comparisons of respelling systems.
- ↑ Monolingual Hebrew dictionaries use pronunciation respelling for words with unusual spelling; for example, the ***Even-Shoshan Dictionary*** respells תְּכַנֵּית as תּוֹכְנִית because the word uses the *kamat* *katan*.
- ↑ For example, **Sergey Ozhegov**'s dictionary adds [hɛ́] in brackets to the French loan-word *пенсне* (*pince-nez*) to indicate that the final (e) does not **iotate** the preceding (h).
- ↑ "In accordance with long-established Czech lexicographical tradition, a modified version of the International Phonetic Alphabet (IPA) is adopted in which letters of the Czech alphabet are employed."^[49]
- ↑ "Segments can usefully be divided into two major categories, consonants and vowels."^[56]
- ↑ They were moved "for presentational convenience [...] because of [their] rarity and the small number of types of sounds which are found there."^[59]
- ↑ "A symbol such as [β], shown on the chart in the position for a voiced bilabial fricative, can also be used to represent a voiced bilabial approximant if needed."^[62]
- ↑ It is traditional to place the tie bar above the letters. It may be placed below to avoid overlap with ascenders or diacritic marks, or simply because it is more legible that way, as in Niesler; Louw; Roux (2005). "Phonetic analysis of Afrikaans, English, Xhosa and Zulu using South African speech databases".^[67]
- ↑ Cf. the /ʷ.../ and /ʱ.../ transcriptions in Ernst-Kurdi, Eszter (2017). "**The Phonology of Mada**" ↗. SIL Yaoundé.
- ↑ E.g. Dolgopolsky, Aaron (2013). *Indo-European Dictionary with Nostratic Etymologies* ↗. Studia Philologica. Rukopisnye pamiatniki Drevnei Rusi.
- ↑ The IPA *Handbook* variously defines the "linking" symbol as marking the "lack of a boundary"^[74] or "absence of a break",^[27] and gives **French liaison** and English **linking r** as examples. It more generally means that the consonant ending one word forms a syllable with the vowel beginning the following word, across an orthographic space. However, the *Handbook* illustration for Croatian uses it to tie atonic **clitics** to tonic words, with no resulting change in implied syllable structure.
- ↑ **^{*a*}****^{*b*}** The global rise and fall arrows come before the affected syllable or prosodic unit, like stress and upstep/downstep. This contrasts with the Chao tone letters (listed below), which most commonly come after. One will occasionally see a horizontal arrow (→) for global level pitch (only dropping due to **downdrift**), e.g. in Julie Barbour (2012) *A Grammar of Neverver*. Additionally, some fonts display the arrows as **emoji** by default, if ︎ is not appended.
- ↑ **^{*a*}****^{*b*}** There is not a one-to-one correspondence between tone/pitch diacritics and tone/pitch letters. When pitch is transcribed with diacritics, the three pitches (é ē è) are taken as the basic levels and are called 'high', 'mid' and 'low'. Contour tones combine only these three and are called (ê) 'high-mid' etc. The more extreme pitches, which do not form contours, are (ě) 'extra-high' and (è) 'extra-low', using **doubled diacritics**. When transcribed with tone letters, however, combinations of all five levels are possible. Thus, (eɭ eɭ eJ) may be called 'high', 'mid' and 'low', with (eɭ eɭ) being 'near-high' and 'near-low', analogous to descriptions of vowel height. In a three-level transcription, (é ē è) are identified with (eɭ eɭ eJ), but in a five-level transcription, (ě è) are identified with (eɭ eJ).^[75]
- ↑ **^{*a*}****^{*b*}** The peaking and dipping diacritics are uncommon and are not illustrated in the IPA *Handbook*. An example is /klä/ 'properly' in *Mian*.
- ↑ Although any combination of tone diacritics is theoretically possible, such as (e) for a falling–rising–falling tone, any others than those illustrated are rare. Double acute – grave is found for an extra-high falling tone in Valentin Vydrin, 2020, *Dan*. In Vossen & Dimmendaal (eds.) *The Oxford Handbook of African Languages*; and double acute – macron is found for an extrahigh-to-mid tone in Roger Blench, 2025, *Tone Systems in the Bantoid Languages*. Neither is supported by Unicode.
- ↑ For example, "**Baleaic**" ↗. *Merriam-Webster.com Dictionary*. Merriam-Webster.
- ↑ Russian and Lithuanian sources and commonly use the character ᳚ VERTICAL SIX DOTS for a less-than-minor break, such as the slight break in list intonation (e.g. the very slight break between digits in a telephone number). ᳚ WIGGLY VERTICAL LINE is used for an unexpected interruption in or a sharp change of intonation.^[78]
- ↑ Maddieson and others have noted that a phonemic/phonetic distinction should be handled by /slash/ or [bracket] delimiters. However, the reversed tone letters remain in use to distinguish tone sandhi from lexical tone when both are phonemic.
- ↑ A work-around sometimes seen when a language has more than one rising or falling tone, and the author wishes to avoid the poorly legible diacritics (ě, ē, ê, è) but does not wish to employ tone letters, is to restrict the generic rising (ě) and falling (è) diacritics to the higher-pitched of the rising and falling tones, say /eʲ/ and /eɿ/, and to resurrect the retired (pre-Kiel) IPA subscript diacritics (e̘) and (e̙) for the lower-pitched rising and falling tones, say /eɹ/ and /eɻ/. When a language has either four or six level tones, the two middle tones are sometimes transcribed as high-mid (è) (non-standard) and low-mid (ê). Non-standard (è) is occasionally seen combined with acute and grave diacritics or with the macron to distinguish contour tones that involve the higher of the two mid tone levels.
- ↑ See for example **Pe Maung Tin** (1924). "bɜːmɪːz". *Le Maître Phonétique*. 2 (39) (5): 4–5.  ↗ JSTOR 44704085 ↗. where five pitch levels are distinguished.
- ↑ The example has changed over the years. In the chart included in the 1999 IPA *Handbook*, it was [ˈ], and since the 2018 revision of the chart it has been [ˠ].
- ↑ Chao did not include tone shapes such as [˥˥], [˨˨], which rise or fall and then level off (or vice versa). Such tone shapes are, however, frequently encountered in the modern literature.
- ↑ In Chao's Sinological convention, a single tone letter (˥) is used for a high tone on a **checked syllable**, and a double tone letter (˥˥) for a high tone on an open syllable. Such redundant doubling is not used in the *Handbook*, where the tones of Cantonese [si˥] 'silk' and [sɨk˥] 'color' are transcribed the same way. If the author wishes to indicate a difference in phonetic or phonemic length, the IPA accomplishes that with the length marks (͡͡ ͡͡͡ ͡͡͡͡) rather than through the tone letters.
- ↑ Sometimes the obsolete transcription (kˈ) (with a turned apostrophe) for weak aspiration vs. (kʰ) for strong aspiration is still seen.
- ↑ E.g. in **Laver 1994**, pp. 559–560
- ↑ The motivation for this may vary. Some authors find the tie bars displeasing but the lack of tie bars confusing (i.e. (č) for /t͡ʃ/ as distinct from /tʃ/), while others simply prefer to have one letter for each segmental phoneme in a language.^[*citation needed*]
- ↑ "At the 1989 Kiel Convention of the IPA, a sub-group was established to draw up recommendations for the transcription of disordered speech."^[105]
- ↑ As in **Afrasianist phonetic notation**. (S) is particularly ambiguous. It has been used for 'stop', 'fricative', 'sibilant', 'sonorant' and 'semivowel'. On the other hand, plosive/stop is frequently abbreviated (P), (S) or (with non-tonal languages) * (T). The illustrations given here use, as much as possible, letters that are capital versions of members of the sets they stand for: IPA [n] is a nasal and (N) is any nasal; [p] is a plosive, [f] a fricative, [s] a sibilant, [l] both a lateral and a liquid, [r] both a rhotic and a resonant, and [x] a click. (ɸ) is an obstruent in Americanist notation, where it stands for [ts]. Alternative wildcards for 'glide', (J) and (W), partly fit this pattern, but are much less common than (G) in English-language sources.
- ↑ In the context of (CRV-) syllables, the (R) is understood to include liquids and glides but to exclude nasals, as in Bennett (2020: 115) 'Click Phonology', in Sands (ed.), *Click Consonants*, Brill
- ↑ {Close vowel} may instead be (U), and (O) may stand for {obstruent}.
- ↑ Or glottal~pharyngeal (H), as in Afrasianist phonetic notation.
- ↑ Somewhat more precisely, (LM) and (MH) are sometimes used for low and high rising tones, and (HM), (ML) for high and low falling tones; occasionally (R) for 'rising' or (F) for 'falling' is seen.
- ↑ For other Turkic languages, (I) may be restricted to {u i} (that is, to *ı* *ı̇*), (U) to *u* *ü*, (A) to *a* *e* (or *a* *ä*), etc.
- ↑ VoQS (V) does not mean phonetic voicing, nor a vowel; for example, in VoQS (V̌) is "nasal voice" (that is, **nasalization**), not a nasal vowel as it would be read in IPA notation.
- ↑ "Diacritics may also be employed to create symbols for phonemes, thus reducing the need to create new letter shapes."^[118]
- ↑ Dedicated letters have been proposed, such as rotated (ḃ) and (ḡ), reversed (ḣ) and (ḡ), or small-capital (ḥ) and (ḡ). Ball, Rahilly & Lowry (2017) *Phonetics for speech pathology*, 3rd edition, Equinox, Sheffield.
- ↑ "The diacritics...can be used to modify the lip or tongue position implied by a vowel symbol."^[120]
- ↑ For example, the IPA *Handbook* lists (p) as "lower-case P" and (x) as "chi."^[125]

References [\[edit \]](#)

Footnotes [\[edit \]](#)

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 - ↑ International Phonetic Association 1999, p. 196
 - ↑ Cf. the notes at the Unicode IPA EXTENSIONS code chart Archived 5 August 2019 at the Wayback Machine as well as blogs by Michael Everson Archived 10 October 2017 at the Wayback Machine and John Wells here Archived 2 June 2019 at the Wayback Machine and here Archived 2 June 2019 at the Wayback Machine.^[*clarification needed*]
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